Complement system I:

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List of Contents:

- Objectives.
- Introduction.
- The complement system.
- Definition.
- Complement characteristics.
- Classical pathway.
- The complement system pathways.
- Summary.

Objectives:

- The students must learn all detail about the definition, characteristics, pathways and biological functions of the complement cascade system.

The complement system:-

- **Definition:-**
- The term (complement) means to supplement (the effect of other components of the immune system, eg. antibody).
- **Complement characteristic:-**
- 1-Complement system consists of a series of proteins that work to "complement" the work of antibodies in destroying microorganism.

The complement system:-

- 2-Compliment proteins are synthesize in liver, tissue macrophages and fibroblasts, this system consists of approximately **20 proteins** that are present in normal human serum.

- 3-It is heat-labile and inactivated by heating serum at 56 c˚ for 30 minutes.

The complement system pathways:-

- There are three pathways for activation of complement cascade:-

- 1-Classical pathway.
- 2-Alternative pathway.
- 3-Lectin pathway.

The complement system:-

- ***All pathways can be divided into three phases:-***
- **1-Initiation phase: -** This phase consists of recognition events.
- **2-Amplification phase: -** The result of this phase is activation of C3
- **3-Membrane attack phase: -** The result of this phase is target cell lyses.

Classical pathway:-

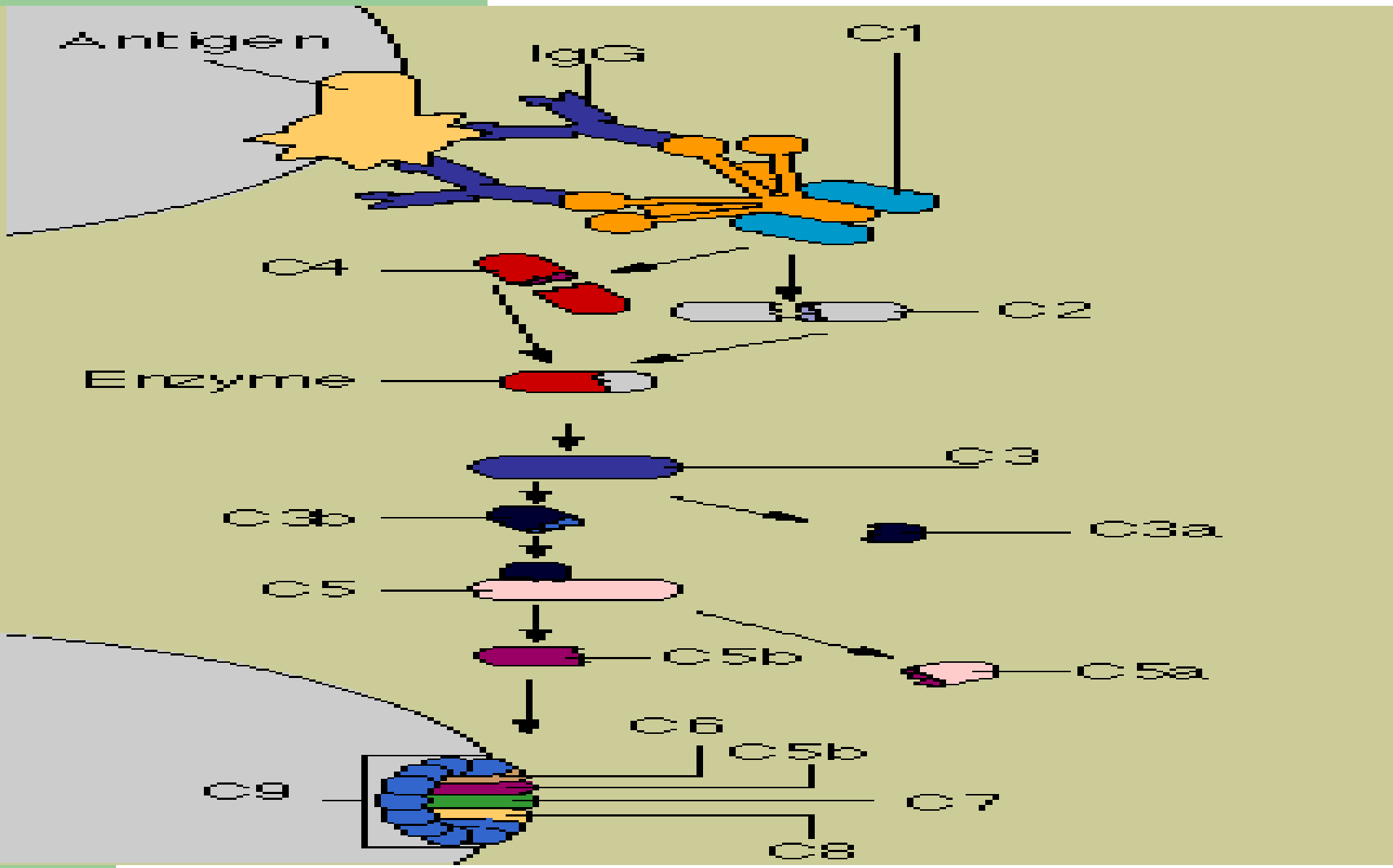
- The classical pathway is activated through antigen-antibody complex.
- Activation follows the binding of complement component C1 to a site on the Fc fragment of the immunoglobulin, CH2 domain on IgG or CH3 on IgM.
- C1 is the recognition unit is a trimolecular complex C1q, C1r and C1s held together by calcium ions.

Classical pathway:-

- Activated C1q leads activation of C1r which in turn activates C1s.
- C1s cleaves C4 into C4a and C4b.
- C4a is an anaphylotoxin fragment stimulates mast cell degranulation and the release of vasoactive amines. C4b bind to cell membranes.

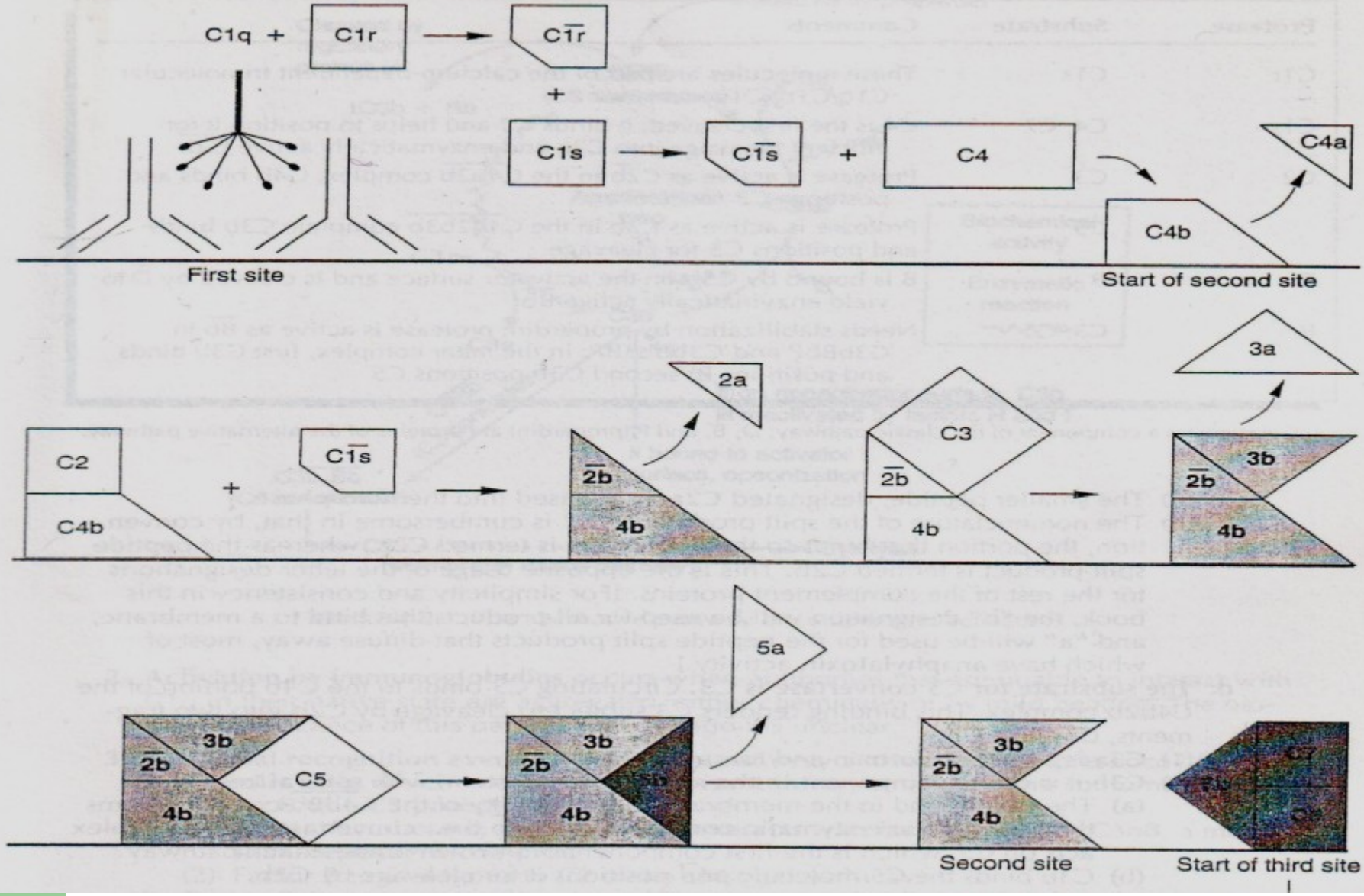
- The next components C2, C3 and C5 will be activated.
- C5a has **anaphylotoxin** activity, while C5b is the first component of the **membrane attack complex.**

Classical pathway:-

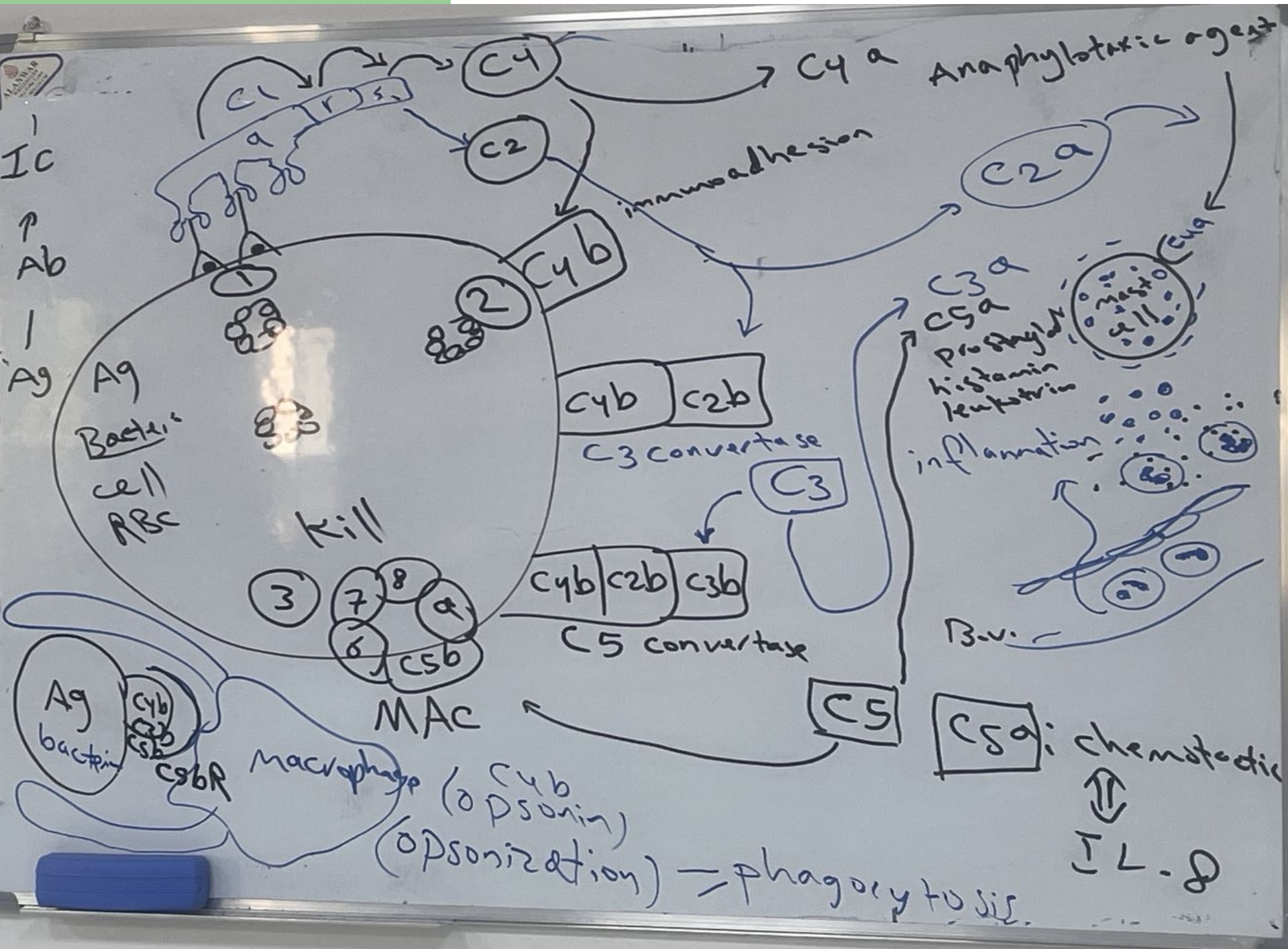


(Internet)

Classical pathway:-



(Hyde, 2000)


C1
r
s
q
C4
C4 a
Anaphylatoxic agent
C2
C2a
immunoadhesion
IC
Ab
Ag
C4b
1
2
Ag
Bacteria
cell
RBC
Kill
C4b C2b
C3 convertase
C3
C3a
C5a
Prostaglandin
histamin
leukotrin
Mast cell
C4a
inflammation
C4b C2b C3b
C5 convertase
3
7
8
9
6
C5b
MAc
C5
C5a
chemotactic
IL-8
Ag
bacteria
C4b
C3b
C3bR
Macrophage
(opsonin)
(opsonization) = phagocytosis

Biologic effects of complement components:-

- **1-Opsonization:-**

  This occurs when activated complement component **coat pathogenic organisms** or immune complexes facilitating the process of phagocytosis.
- **2-Chemotaxis:-**
- C5a and the C567 complex **attract neutrophils**. C5a also enhances the **adhesiveness** of neutrophils to the endothelium.

Biologic effects of complement components:-

- **3-Anaphylotoxin:-**

  C3a,C4a and C5a cause **degranulation of mast cells** with release of mediators as **histamine** leading to increased vascular permeability and smooth muscle contraction.

Biologic effects of complement components:-

- **4-Cytolysis:-**

  In the final stage of the complement cascade, the membrane of target cells(erythrocytes, bacteria and tumor cells) are attacked by **membrane attack complex(MAC)** leading to cell destruction by entry of water and electrolytes into the cell.

Principle inflammatory activities of activated

Component	Activity
C3a, C5a	Anaphylatoxin—releases histamine, and other vasoactive compounds from basophils and mast cells, increasing capillary permeability
C3b, iC3b, C4b	Immune adherence and opsonization—binds antigen—antibody complexes to membranes of macrophages and neutrophils, enhancing phagocytosis; also binds complexes to erythrocytes, facilitating removal by the liver and spleen
C5a	Chemotaxis and chemokinesis—attracts phagocytic cells to sites of inflammation and increases their overall activity
C8, C9	Membrane damage—transmembrane channels form, permitting flux of cytoplasmic constituents. Mammalian cells swell and burst; bacterial cells become leaky and lose vital intracellular metabolites, but usually do not burst
Ba	Neutrophil chemotaxis
Bb	Macrophage activation—causes macrophages to adhere to surfaces and spread on them

(Hyde, 2000)

Summary

The complement system consists of a series of proteins that work to "complement" the work of antibodies in destroying microorganism.

There are three pathways for activation of complement cascade, classical , alternative and lectin pathways. The system is undergo three phases Initiation phase, amplification phase and membrane attack phase.

References

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Thanks a lot